



Project Proposal

Data analysis and visualization

Phase 1

Topic : Lahore Weather Analytics Lakehouse Pipeline

Bds - 5B

Group members

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Phase 1 Project Proposal

Lahore Weather Analytics Lakehouse Pipeline

Domain: Climate & Meteorological Analytics

Data Architecture: Medallion Architecture (Bronze → Silver → Gold) via Apache Spark

GitHub Repository:

<https://github.com/Syedafaryalzehra/Lahore-Weather-Analytics-Lakehouse-Pipeline/tree/main>

1. Domain & Source Identification

Project Concept

This project focuses on building an automated, distributed data pipeline for Weather and Climate Analytics for the city of Lahore, Pakistan. Weather data is one of the most consistently generated, publicly available, and analytically rich datasets in the world. By ingesting decades of hourly meteorological readings for Lahore, this pipeline enables analysts, urban planners, and researchers to identify long-term climate trends, seasonal patterns, extreme weather events, and year-over-year changes in temperature, rainfall, and wind conditions.

Data Source

We are using the Open-Meteo Historical Weather API — a completely free, open-source meteorological API that requires no API key, no signup, and no authentication of any kind.

- API Name: Open-Meteo Archive API
- Base URL: <https://archive-api.open-meteo.com/v1/archive>
- Data Format: JSON responses containing hourly weather measurement arrays
- Source Access: Fully public, no registration required
- Cost: 100% Free, no credit card needed
- Location: Lahore, Pakistan (Latitude: 31.5204, Longitude: 74.3587)

Ingestion Pattern & Dual-Load Capability

- **Full Load:**
Using the archive endpoint with start_date=2000-01-01 and end_date=2024-12-31, we extract 24 years of hourly weather data for Lahore across 18 meteorological variables. This one-time pull establishes our complete historical baseline dataset of approximately 210,000 hourly records.
- **Incremental Load:** Every day, the pipeline fetches only the previous day's 24 hourly readings using past_days=1. Since Open-Meteo finalizes weather readings 24 hours

after observation, this daily pattern always captures confirmed, complete data. Only new records are appended — no reprocessing of historical data.

2. Data Samples & Volume

Sample Files

Representative raw sample files have been uploaded to our GitHub repository under the `/data/` directory. Note: GitHub sample files represent a small slice of the dataset for review purposes. The actual production pipeline ingests the full 24-year date range.

- `data/full_load/full_load_sample.json`: 7 days of hourly weather data for Lahore (January 1–7, 2024) across 18 meteorological variables, representing the structure of the full historical baseline load.
- `data/incremental_load/incremental_load_sample.json`: 1 day of hourly readings (January 8, 2024) representing a single daily incremental batch as it would arrive in the live pipeline.

Volume & Frequency Estimation

- Initial Full Load Baseline: Approximately 200–300 MB of JSON data covering 24 years (2000–2024) of hourly readings across 18 variables, totalling approximately 210,000 hourly records. This volume provides a realistic payload that justifies Apache Spark's distributed partitioning strategies while remaining safely within Databricks Community Edition free compute limits.
- Incremental Load Frequency: Daily batch ingestion of 24 hourly records per day, approximately 50–100 KB per daily execution run.
- Monthly Accumulation: Approximately 1–2 MB of new data added per month of pipeline operation.

3. Security & Compliance

Identification of PII and Sensitive Data

After a thorough audit of all fields returned by the Open-Meteo API, we confirm that this dataset contains absolutely zero Personally Identifiable Information (PII).

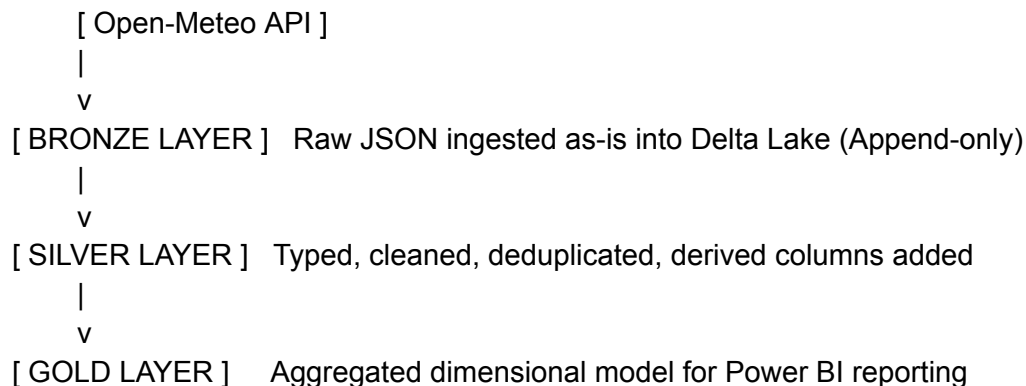
- Weather measurement fields (temperature, precipitation, wind speed, humidity, etc.): Pure environmental sensor measurements with no connection to any individual person.
- Timestamps: Fixed hourly time slots representing measurement intervals, not tied to any individual's activity.
- Location coordinates: Fixed city-level coordinates for Lahore, not personal location data.
- Soil and atmospheric variables: Environmental sensor readings only.
- Names, email addresses, phone numbers, IP addresses, financial details: Not present. The Open-Meteo API by design never returns any personal data in any endpoint.

Handling Strategy

Since no PII exists in this dataset at any point, no hashing, masking, or anonymization steps are required. However, the following data governance practices will still be applied:

1. Timestamp Standardization: All time values will be converted from Asia/Karachi timezone to UTC during the Bronze-to-Silver transition to ensure consistency across the pipeline.
 2. Schema Enforcement: Delta Lake schema enforcement will be enabled to automatically reject any unexpected new fields that could be introduced in future Open-Meteo API version updates.
 3. Data Quality Quarantine: Rows with null values or physically impossible readings (for example, temperature above 60 degrees C or below -10 degrees C for Lahore) will be flagged and written to a separate `_quarantine` table rather than silently dropped, preserving data lineage.
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4. High-Level Medallion Data Modeling



Bronze Layer (Raw Ingestion)

- Ingests raw JSON responses as-is into Delta Lake, append-only, no transformations.
- Adds metadata columns: `ingestion_timestamp`, `load_type`, `batch_date`.
- Partitioned by `ingestion_date`. Malformed records written to `_quarantine` table.

Silver Layer (Cleansed & Typed)

- Type Casting: All fields cast to proper Spark types — TimestampType, DoubleType, IntegerType.
- Timestamp Normalization: Converted to UTC. Year, month, day, hour extracted as separate columns.
- Null Handling: Forward-fill for isolated nulls. Rows with 3+ consecutive null hours quarantined.
- Duplicate Removal: Deduplicated on timestamp as primary key.
- Derived Columns: season (Spring/Summer/Monsoon/Winter), time_of_day, heat_index added.

Gold Layer (Aggregations & Star Schema)

- FactHourlyWeather: One record per hour, all cleaned measurements. Main fact table.
- AggMonthlyClimate: avg/max/min temperature, total precipitation, avg wind per month per year.
- AggYearlyExtremes: Hottest day, coldest day, wettest day, highest wind per year.
- DimDate: Standard date dimension for Power BI time-intelligence queries.

Section 5 — Business Intelligence & Dashboards

The final Power BI dashboard will answer three core questions:

1. How has Lahore's temperature changed over the last 24 years?
2. Which months and seasons receive the most rainfall?
3. What are the most extreme weather events recorded in Lahore?

Visual Charts & Key Metrics

- Annual Temperature Trend Line (2000–2024): Line chart showing average annual temperature over 24 years. Highlights long-term warming trends and climate change patterns for Lahore.
- Monthly Rainfall Bar Chart: Clustered bar chart showing average precipitation per month across all years, color coded by season with Monsoon months highlighted. Useful for flood risk and agricultural planning.
- Extreme Weather KPI Cards + Heatmap: Four KPI cards showing Hottest Day, Coldest Day, Wettest Day, and Highest Wind Speed ever recorded. Accompanied by a calendar heatmap of daily maximum temperatures across all years.

6. Engineering Setup & FinOps

Version Control & Repository Structure

All source code, notebooks, sample data, and documentation are hosted on GitHub:

Repository

Link: <https://github.com/Syedafaryalzehra/Lahore-Weather-Analytics-Lakehouse-Pipeline/tree/main>

- Full load sample: data/full_load/full_load_sample.json
- Incremental load sample: data/incremental_load/incremental_load_sample.json

FinOps & Cost Awareness

All infrastructure used in this project is completely free:

- Distributed Processing: Databricks Community Edition
- Storage: Delta Lake with Snappy compression
- API: Open-Meteo — no cost, no key, no limits
- Dashboard: Power BI Desktop (Free)
- Version Control: GitHub (Free)
- Total Cost: \$0.00